Accurate Modelling of Liquid Crystal Metasurfaces for Spectrum Modulation

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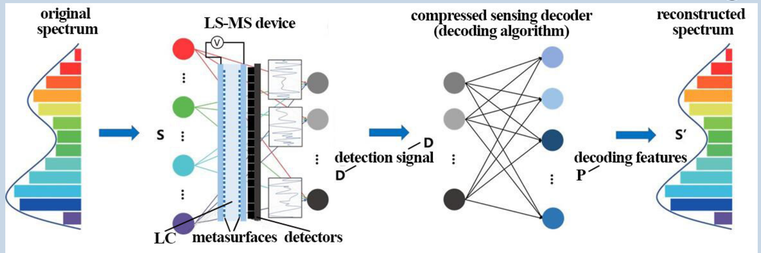
Background

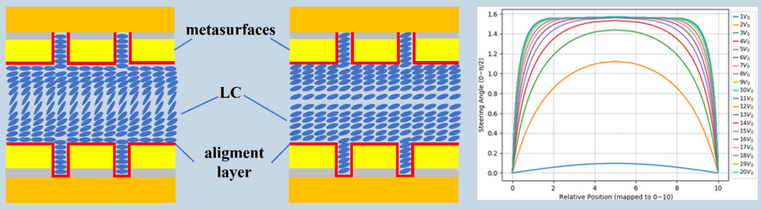
Optical metasurfaces control light's amplitude, phase, and polarisation through spatially arranged meta-atoms. Integrating liquid crystals (LCs) with metasurfaces enables tunable optical responses via electrically controlled birefringence. Such liquid crystal metasurfaces (LCMs) combine high endurance with precise modulation, making accurate optical simulation essential for device design.



Influencing Factors

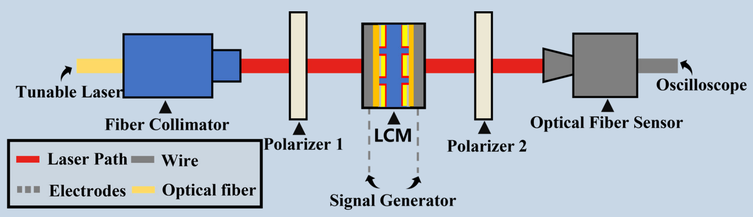
The liquid crystal orientation is primarily influenced by two categories of effects:

- **Layer effects:** The ITO, Cr, and alignment layers jointly modify the local electric field distribution and surface anchoring conditions.
- **Non-uniform LC distribution:** In the existing approach, the orientation of LC molecules is assumed to be spatially uniform. But this assumption was incorrect.



Experimental Setup

A polarized tunable laser probes the voltage-driven LCM device, with transmitted light collected and analyzed after precise polarization alignment.



Modelling of LC

The LC orientation $\phi(z)$ under applied voltage V was obtained using the extended Oseen–Frank model, accounting for elastic and dielectric effects:

$$\frac{V}{V_0} = \frac{1}{\pi} \times \int_0^{\phi_m} \sqrt{\frac{(1+\kappa \sin^2\phi)(\sin^2\phi_m - \sin^2\phi)}{1+\kappa\sin^2\phi_m}}\, d\phi,$$

$$V_0 = \pi\sqrt{\frac{k_{11}}{\varepsilon_0(\varepsilon_\parallel - \varepsilon_\perp)}}.$$

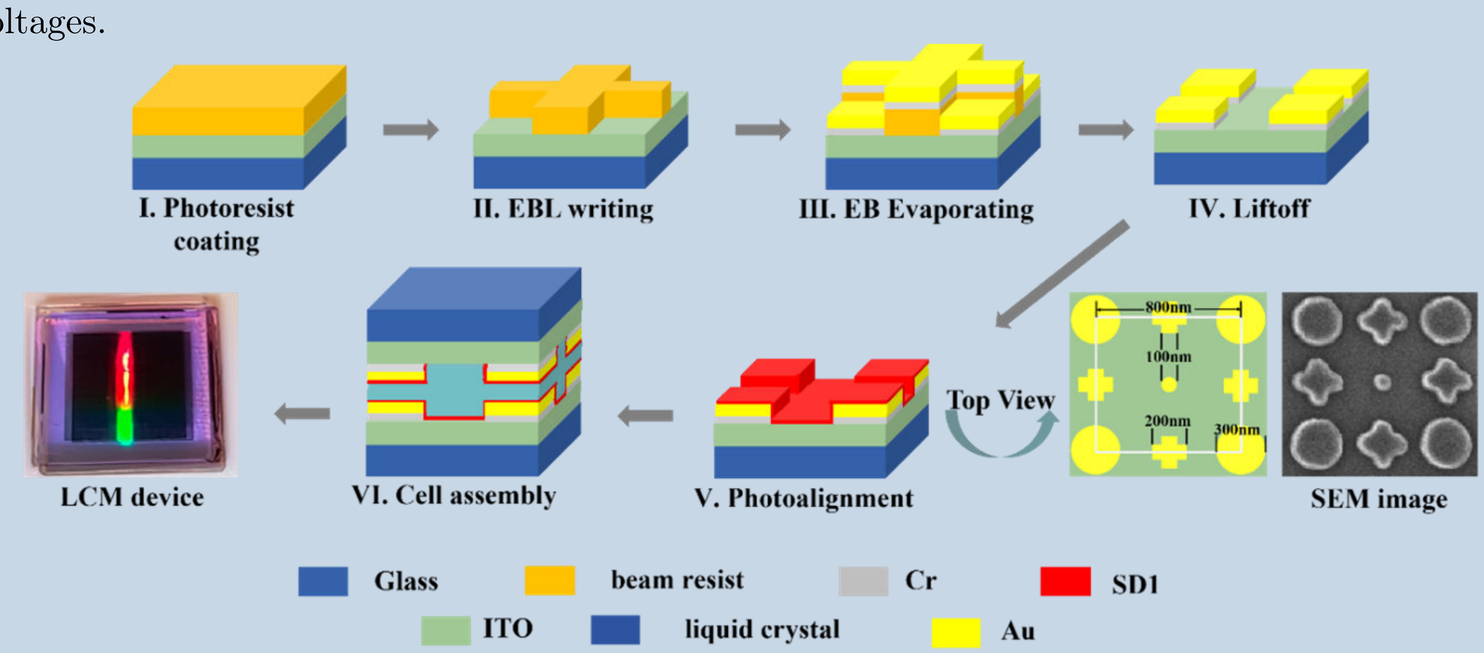
where $\kappa = (k_{33} - k_{11})/k_{11}$. The maximum tilt $\phi_m(V)$ was solved iteratively via the Newton–Raphson method:

$$\phi_m^{(n+1)} = \phi_m^{(n)} - \frac{V(\phi_m^{(n)}) - V_{\text{applied}}}{dV/d\phi_m|_{\phi_m^{(n)}}}.$$

Device structure and fabrication

The fig following shows the structure of an LCM device used to validate the proposed model in this study and its fabrication process. This device consists of an LC layer sandwiched between a pair of glass substrates with embedded gold metasurface structures. It intends to exploit the sensitivity of the metasurface and was designed to generate a series of distinct spectral filtering functions as the driving voltage applied to the LC layer changed.

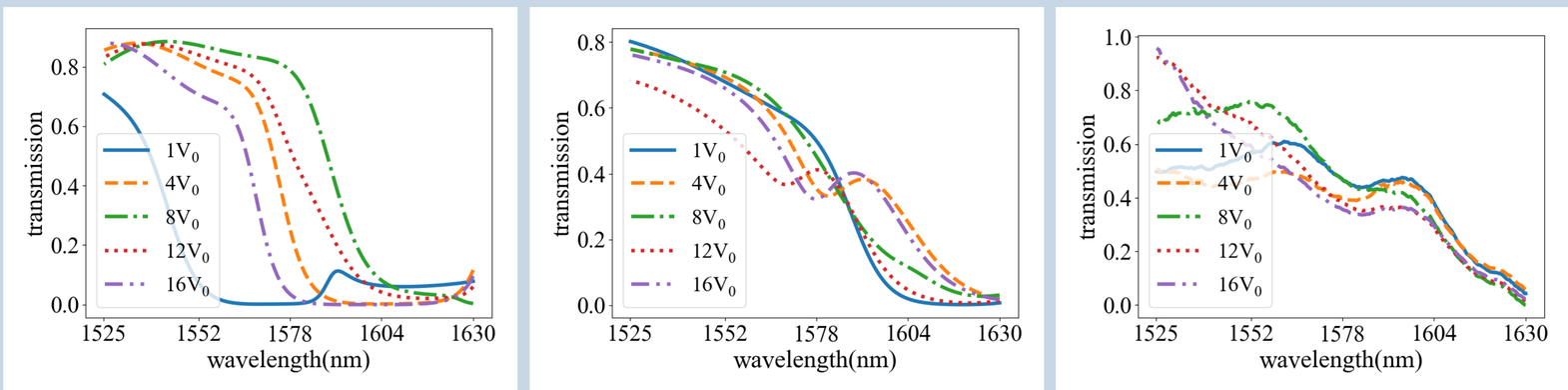
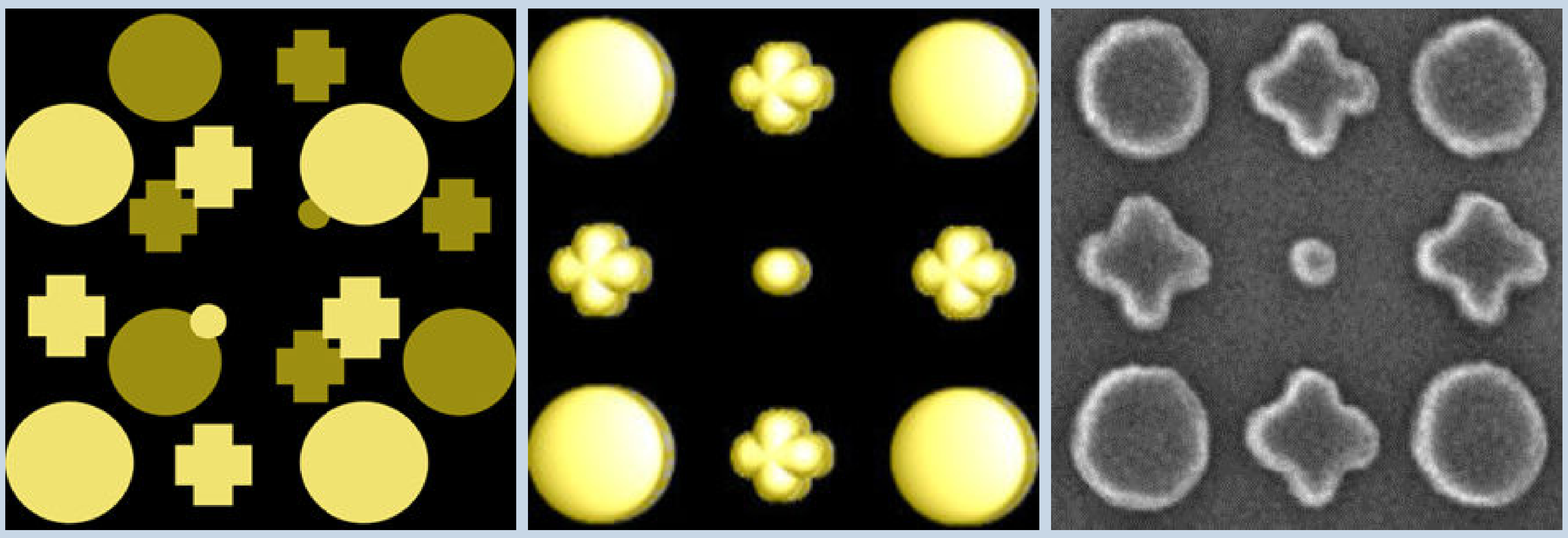
The metasurface structure features a C4-symmetric square gold unit with an 800 nm periodicity in both horizontal and vertical directions. Each unit contains four larger cylinders at the corners, a smaller cylinder in the centre, and four crosses along the borders. The thickness of these structures is 100 nm. The combination of these structures can excite strong localised surface plasmon resonances (LSPR) at different wavelengths, leading to more complex spectral responses under different driving voltages.



Results and Discussion

We developed a customised finite-difference time-domain (FDTD) script to simulate the spectral responses of the proposed LCM device to the linearly polarised beam within the near-infrared wavelength range.

We simulated the LCM's optical responses under different driving voltages. The left figure shows the results obtained by existing model, in which the orientation of the LC molecules was assumed to be spatially uniform. The middle figure shows the results from our model in this work. The right figure shows the experimental results. These results demonstrate that our model achieves significant improvements compared to the baseline.



The remaining discrepancy between our model and the experiment results can be attributed to the fabrication and assembly error of the LCM device under investigation in this work. The 3D profile of the fabricated metasurface substrate was deviated from the design. The material loss occurred during the lift-off process in the fabrication, especially at sharp corners. This led to unintended 3D rounding that altered the structural geometry of the metasurface. In addition, the misalignment between the two metasurface substrates inevitably occurred during the encapsulation of the LC materials. The left figure below is the top-view illustration of the two metasurfaces layers exhibiting misalignment, with an offset of 0.25 cycles, The middle figure is the top view of metasurfaces prongs after abrasion, and the right figure is the scanning electron microscopy (SEM) image of the fabricated metasurfaces. This, to some extent, proves that our conjecture was right.